

Gioia Zheng

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Profile

Undergraduate at Sapienza University of Rome working on retrieval and evaluation systems for retrieval-augmented generation. Focus on full-pipeline construction (sparse + dense retrieval, reranking, generation), paired-bootstrap evaluation methodology, and reproducibility scaffolding.

Education

Sapienza University of Rome

Sep 2022 – **TODO** *expected graduation*

B.Sc. in Applied Computer Science and Artificial Intelligence

Weighted average: **TODO** *e.g. 28/30*

Relevant coursework: **TODO** *4–6 items, e.g. Machine Learning, Algorithms, Information Retrieval, Statistics, Distributed Systems*

Thesis: **TODO** *topic if defined; otherwise omit this line*

Experience

NLP Algorithm Engineer Intern

Feb 2026 – May 2026

Microsoft — Natural Language Computing (Remote)

- **Token-F1 0.197 → 0.368** on MS MARCO dev/small (6,980 queries) by adding a cross-encoder reranker over dense top-100; 95% paired-bootstrap CIs on all four surface-form Δ s strictly above 0.
- Built the end-to-end pipeline: BM25 and dense (SBERT/FAISS) retrieval, cross-encoder reranking, T5 generation, paired-bootstrap evaluation harness.
- +0.173 **DistilBERT-BERTScore Δ** on 3,000 paired examples ruled out a surface-form artefact; deterministic failure taxonomy attributed **~90% of regressions to generator-side truncation**, not retrieval failure.
- **NLI-entailment Δ sign reversal** (−0.145) — the only metric in the pipeline whose paired Δ reverses sign — flagged as evidence for generator-capacity follow-up rather than decode-budget tuning.
- Reproducibility scaffolding: per-run manifest (git SHA, config hash, dependency-file hashes, environment fingerprint), pinned lockfile, 286-test pytest suite, CI on every push.

Projects

Handwritten OCR pipeline

github.com/GioiaZheng/handwritten-ocr-system

PyTorch — CNN + BiLSTM + CTC

- Variable-length transcription pipeline on the IAM handwriting dataset: CNN feature extractor, BiLSTM encoder, CTC decoder.
- Train / dev / test split with held-out evaluation reporting CER and WER; numbers comparable across architecture sweeps.
- End-to-end training and inference scripts; thin web front-end for interactive evaluation.

Chat service

github.com/GioiaZheng/go-chat-system

Go (concurrent backend) + Vue + SQLite

- REST API documented under OpenAPI; Vue frontend embedded into the Go binary at build time for single-artifact deployment.
- Concurrent request handling with SQLite persistence; CI on every push (gofmt, go-vet, tests, OpenAPI lint, frontend build).

TODO *third project: retrieval / evaluation / infra-oriented*

TODO *candidate scopes — a small FAISS demo with a published HF dataset card; a Pyserini-comparable BM25 baseline; a small open-source PR to ir-measures / pyserini / sentence-transformers*

Open-source / Writing

TODO *one or two entries — e.g. a methodology blog post on surface-form vs. NLI-entailment disagreement in RAG evaluation; a HuggingFace dataset card; an issue or PR against pyserini / ir-measures / BEIR / sentence-transformers*

Skills

Primary

Python, PyTorch, BM25, FAISS, sentence-transformers, cross-encoder reranking, paired-bootstrap evaluation, NLI grounding, BERTScore

Working

Go (concurrent backends), Docker, Linux, GitHub Actions CI, OpenAPI, SQLite

Familiar

JavaScript / React Native, Vue

Languages

English (Professional), Italian (Bilingual), Chinese (Native), German (Beginner)